

PARKTOWN BOYS' HIGH SCHOOL



Department of Mathematics

Mid-Year Examination

Paper 1

4 June 2025

Subject	Mathematics
Grade	12
Marks	150
Time	3 hours

Examiner

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Moderator

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INSTRUCTIONS

Read the following instructions carefully before answering the questions.

1. This question paper consists of 11 questions, 9 pages and a formula sheet. Please ensure that your question paper is complete. You may detach the formula sheet for your own reference.
2. Answer ALL the questions in the answer book provided. Please note clearly where the additional space is being used.
3. Clearly show ALL calculations, diagrams, graphs, etc. that you have used in determining your answers.
4. Answers only will NOT necessarily be awarded full marks.
5. You may use an approved scientific calculator (non-programmable and non-graphical), unless stated otherwise.
6. If necessary, round off answers to TWO decimal places, unless stated otherwise.
7. Diagrams are NOT necessarily drawn to scale.
8. Use the mark allocation to guide the length of your solution.
9. Write neatly and legibly.

QUESTION 1

1.1 Solve for x :

1.1.1 $(x-1)(x+2) = 4$ (3)

1.1.2 $2x^2 + 5x = 1$ (3)

1.1.3 $x^2 - 4x - 21 \geq 0$ (4)

1.1.4 $3^{2x} + 2 \cdot 3^x - 15 = 0$ (4)

1.1.5 $\sqrt{x-2} + \frac{12}{\sqrt{x-2}} = 7$ (5)

1.2 Solve for x and y simultaneously:

$$\begin{aligned} 2x + y &= 1 \\ x^2 - 3xy + y^2 + 2y &= 25 \end{aligned} \quad (5)$$

1.3 Show that for all positive integer values of n , $2 \cdot 7^n - 7^{n+1} + 7^{n+2}$ is an even number. (3)

[27]

QUESTION 2

It is given that $5 ; 7 ; 9 ; 11 ; \dots$ is the sequence of first differences of a quadratic sequence.

2.1 Determine the sum of 70 terms of the sequence of first differences. (3)

2.2 If the third term of the quadratic sequence is 14, find the general term of the quadratic sequence. (5)

2.3 Determine which term of the quadratic sequence equals 959. (4)

[12]

QUESTION 3

3.1 The second term of a geometric sequence is 6 and the seventh term is 192.

3.1.1 Determine the first term and the constant ratio of this sequence. (4)

3.1.2 Determine the minimum value of m for: $\sum_{k=1}^m 3 \cdot 2^{k-1} > 1\,000\,000$ (5)

3.2 If the first term of a geometric series is $2t - 4$ and the second term is $t^2 - 5t + 6$.

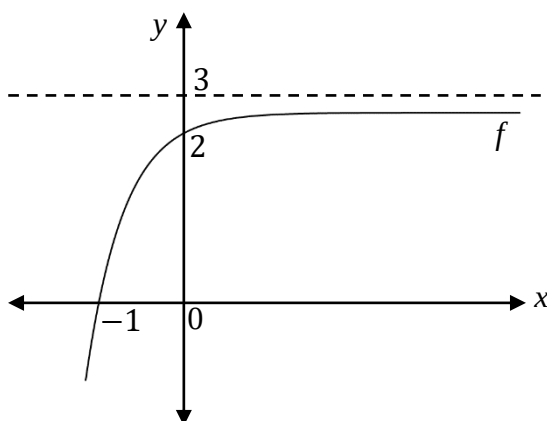
3.2.1 For which values of t will the series converge? (4)

3.2.2 For what value of t will the sum to infinity of the series be 0? (3)

[16]

QUESTION 4

4.1 The graph of $f(x) = -b^x + c$ is drawn on a set of axes below:



4.1.1 Write down the value of c . (1)

4.1.2 Determine the value of b . (2)

4.1.3 If f is shifted 1 unit to the right and 4 units downwards to form g , determine the equation of g in the form $g(x) = a \cdot b^x + c$. (3)

4.2 Given: $g(x) = \log_{\frac{1}{2}} x$

4.2.1 Determine the equation of g^{-1} in the form $y = \dots$ (2)

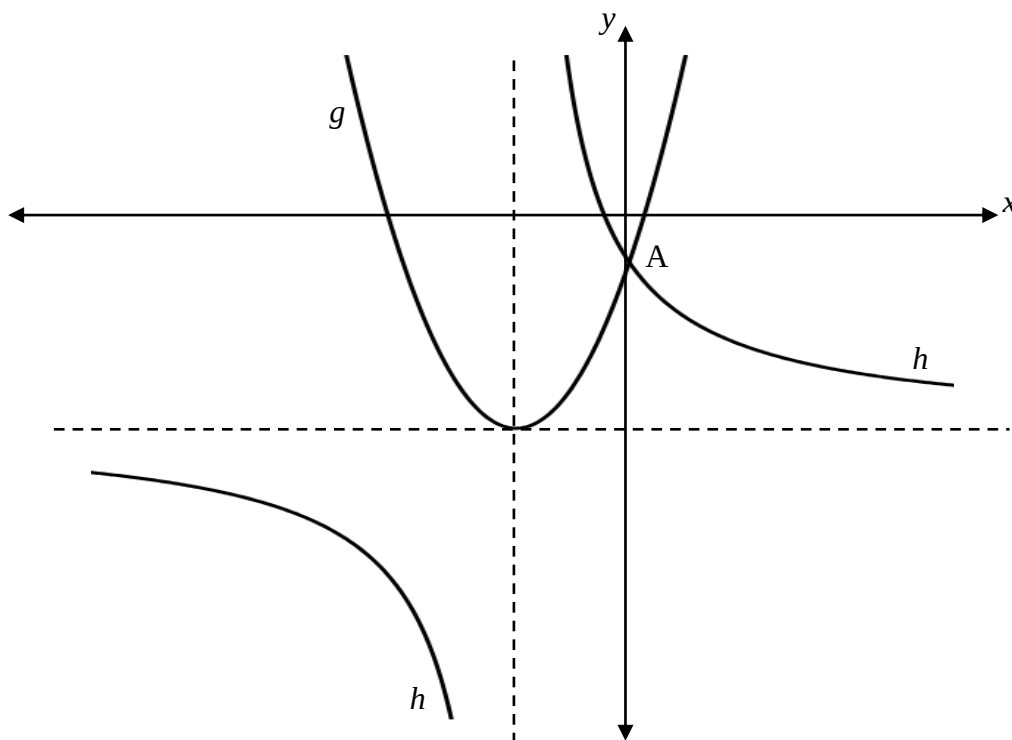
4.2.2 Sketch g and g^{-1} on one set of axes, showing the axis of symmetry. (5)

4.2.3 State the range of g^{-1} . (1)

[14]

QUESTION 5

The graphs of $g(x) = a(x + p)^2 + q$ and $h(x) = \frac{2}{x+1} - 3$ are sketched below. A is the y-intercept of g and h . The horizontal asymptote of h is also a tangent to g at the turning point of g .



- 5.1 Write down the equation of the vertical asymptote of h . (1)
- 5.2 Determine the coordinates of A. (2)
- 5.3 Determine the value of a . (3)
- 5.4 j is one of the axes of symmetry of h where $j'(x) < 0$. Determine the equation of j . (2)
- 5.5 For which values of k will $g(x) + k = 0$ have two negative roots? (2)
- 5.6 Find the values of x for which $g'(x) \cdot h'(x) > 0$. (2)
- 5.7 If $f(x) = 2x + m$, determine the value of m for which f is a tangent to g . (6)

[18]

QUESTION 6

A function f is given by $f(x) = x^2 - 3$. Determine the value(s) of x for which $f'(x) = f^{-1}(x)$, where $x > 0$. (6)

[6]

QUESTION 7

7.1 An owner of a company buys a machine priced at R120 000. The value of the machine depreciates at a rate of 9% per annum according to the reducing balance method.

7.1.1 Calculate the scrap value of the machine by the end of 5 years. (2)

7.1.2 After five years, the machine needs to be replaced. During the course of this time, inflation remained constant at 7% per annum. Determine the cost of the new machine by the end of 5 years. (2)

7.1.3 The company sells the machine at scrap value after 5 years and uses it to purchase a new machine. A sinking fund is set up to finance the balance, such that equal monthly instalments must be paid. Interest on this fund is at the rate of 8,5% per annum, compounded monthly. The first payment will be made immediately and the last payment will be made at the end of the 5 year period. Calculate the value of the monthly payment that needs to be made into the sinking fund. (4)

7.2 A business purchases a property for R1 500 000. After paying a deposit, it takes out a loan of R1 275 000 for the remaining amount at an interest rate of 9,2% per annum, compounded monthly over a period of 20 years. The monthly instalment on the loan is R11 636,02.

7.2.1 Calculate the outstanding balance on the loan after 7 years. (4)

7.2.2 After 7 years, the business misses 5 consecutive payments due to financial difficulties. Thereafter, it continues making monthly payments into the loan account until the end of the 20 year period. Calculate the value of the new monthly instalment required to settle the loan. (4)

[16]

QUESTION 8

8.1 If $g(x) = x^3$, determine $g'(x)$ from first principles. (5)

8.2 Determine:

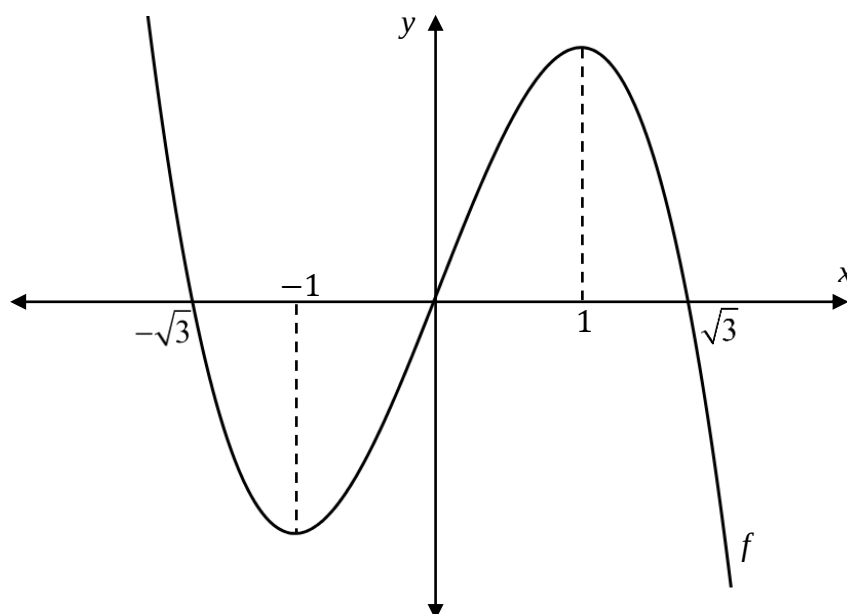
8.2.1 $f'(x)$ if $f(x) = (x-3)(2x+1)$ (3)

8.2.2 $\frac{d}{dx} \left(\sqrt[3]{x^5} - \frac{4}{x^4} \right)$ (3)

[11]

QUESTION 9

The graph of $f(x) = -2x^3 + bx^2 + cx + d$, a cubic function passing through the origin, is drawn below. The x -intercepts are at $\pm\sqrt{3}$. The x -coordinates of the stationary points of f are -1 and 1 .



9.1 Determine for which values of x f decreases. (2)

9.2 Write down the x -coordinates of the x -intercepts of f' . (2)

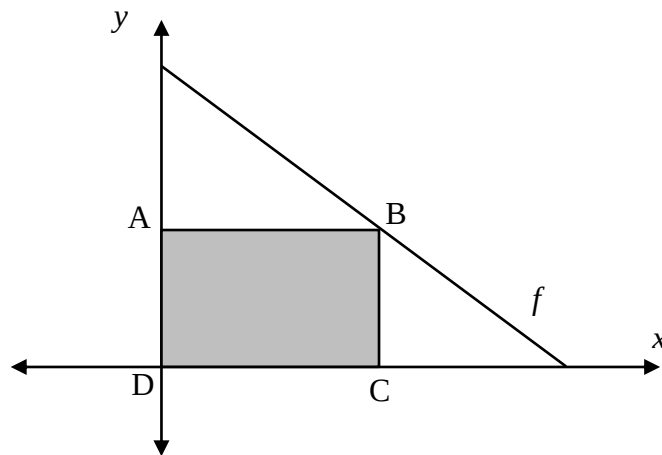
9.3 For which values of x is f concave up? (2)

9.4 Find the equation of f . (4)

[10]

QUESTION 10

The rectangle ABCD touches the graph $f(x) = -3x + 12$ at B.



10.1 Show that the area of ABCD can be expressed as $A(x) = -3x^2 + 12x$. (2)

10.2 Determine the maximum area of rectangle ABCD. (4)

[6]

QUESTION 11

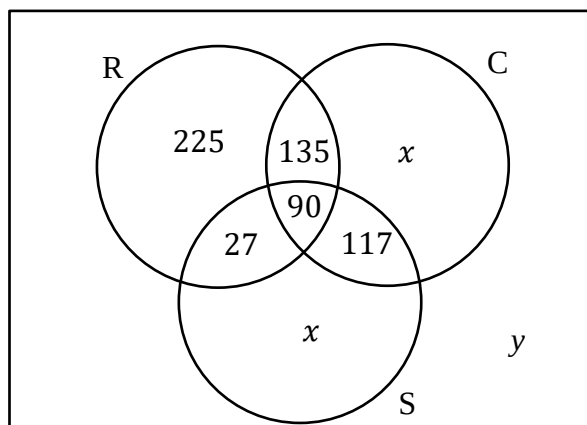
11.1 C and D are independent events. It is further given that $P(C \text{ and } D) = 0,3$ and $P(D) = 0,5$. Calculate:

11.1.1 $P(\text{only } C)$. (4)

11.1.2 $P(\text{not } C \text{ or not } D)$. (2)

11.2 A survey is done on the sport in which matriculants participate. The survey considered Rugby (R), Cricket (C) and Swimming (S). 900 matric learners were surveyed and the following results were obtained and represented in the Venn diagram below:

- 225 learners do rugby only
- 135 learners do rugby and cricket, but not swimming
- 117 learners do cricket and swimming, but not rugby
- 27 learners do rugby and swimming, but not cricket
- 630 learners participate in at least one of the sports
- 90 learners participate in all three sports
- x learners participate in cricket only and the same number of learners participate in swimming only



11.2.1 Determine the probability that a randomly selected learner participates in all three activities. (1)

11.2.2 Determine the value of x . (2)

11.2.3 Determine the probability that a randomly selected learner participates in none of the activities. (3)

11.2.4 Determine the probability, as a percentage, that a randomly selected learner participates in rugby or cricket. (2)

[14]